

Dr. Rohitkumar R. Upadhyay

Applied Cryptographer, QNu Labs



Contact Information

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Profile Summary

Ph.D. in Cryptography specializing in the formal design and security analysis of privacy-preserving protocols (FHE, MPC, ZKPs) and post-quantum algorithms (PQC). Expert in bridging mathematical complexity theory with high-assurance implementations, with a focus on hardening cryptographic primitives against side-channel leakages and optimizing protocol efficiency through rigorous algorithmic modeling.

Research Interests

Lattice-based Cryptography, Efficient FHE/MPC, Verifiable Computation (SNARKs), Side-Channel Analysis (SCA), and Quantum-Resistant Protocol Design.

Education

- 2022 – 2025 **Ph.D. in Mathematics**, [MNNIT Allahabad](#), India
○ Thesis: A Study of Secure Computation using Post-Quantum Cryptography
○ Advisor: [Dr. Sahadeo Padhye](#) **CGPA: 9.5/10**
○ Key Contributions:
 - Proposed efficient **Dynamic Multi-Party FHE** frameworks based on **GSW** and **FHEW**, enabling scalable secure computation in decentralized environments.
 - Developed a multi-key FHE scheme over integers with batching mechanisms for parallel operations in large-scale MPC.
 - Designed **FHE-Pro**, integrating Proxy Re-encryption to enable non-interactive key generation and reduced communication complexity.
 - Optimized homomorphic multiplication using **bit decomposition**, reducing computational complexity from quadratic to linear for structured data and PPML.
 - Formalized **SNARK-driven verifiable FHE** mechanisms to ensure result integrity and privacy in outsourced cloud computations.
 - Investigated **VOLE-in-the-Head** for constructing efficient post-quantum digital signatures.
- 2017 – 2019 **M.Sc. in Mathematics**, [University of Mumbai](#), India, CGPA: 7.01/10
- 2014 – 2017 **B.Sc. in Mathematics**, [NES Ratnam Degree College, Mumbai](#), India, CGPA: 6.62/7

Experience

- Apr 2026 – **Applied Cryptographer**, [QNu Labs](#), Delhi, India
Present
- Jun 2025 – **Cryptography Analyst**, [QNu Labs](#), Bangalore, India
Mar 2026
- Developing **algorithmic optimizations** and **formal security analyses** for constant-time ML-KEM and ML-DSA to ensure resistance against physical leakages.
 - Formulating **theoretical leakage models** and designing robust masking and blinding countermeasures for post-quantum cryptographic primitives.
 - **Designing and formally modeling** efficient MPC and FHE-based protocols for secure data sharing, focusing on reducing communication and improving scalability.
 - Evaluating NIST PQC standards and global benchmarks to inform internal transition strategies.
- Sep 2024 – **Trimester Laureate**, *Institut Henri Poincaré (IHP)*, Paris, France
Dec 2024
- Selected as one of three global laureates for the IHP Trimester on Post-Quantum Algebraic Cryptography; worked with Prof. [Ludovic Perret](#) on MPC-in-the-Head-based signatures.
- 2019 – 2022 **Assistant Professor**, [KET's Vinayak Ganesh Vaze College](#), Mumbai, India
- Taught undergraduate courses including Topology, Number Theory, and Cryptography.

Technical Skills

- PPC Libraries **OpenFHE, SEAL, Lattigo, libsnark, Circom, MP-SPDZ**
- PQC & SCA **PQClean, liboqs, ChipWhisperer**, Constant-time coding, Masking/Blinding
- Programming **C/C++ (Advanced), Python (Advanced - SAGE, NumPy), Go (Intermediate)**

Research Publications

- 2025 **Upadhyay, R.R., Padhye, S., Sahu, R.A., Saraswat, V.** *[Towards Privacy and Integrity: SNARK-Driven Verifiable FHE for Outsourced Computation](#)*. In: **ACNS 2025 Workshops**. Springer LNCS.
- 2025 **Upadhyay, R.R., Padhye, S., Saraswat, V.** *[A Non-Interactive and Dynamic Multi-Party FHE Scheme Based on FHEW](#)*. **Peer-to-Peer Networking and Applications**. (SCIE).
- 2025 **Upadhyay, R.R., Padhye, S.** *[A Non-Interactive and Secure Dynamic Multi-Party FHE Scheme](#)*. **International Journal of Information Security**. (SCIE).
- 2025 **Kumar, R., Dabral, A., Upadhyay, R.R.** *[Cryptanalysis of a Lattice-Based Blind Ring Signature Scheme](#)*. **National Academy Science Letters**. (SCIE).
- 2025 **Upadhyay, R.R., Padhye, S.** *[SoK: Digital Signatures from VOLEith](#)*. **Human-2025 Symposium**. (Best Paper Award).
- 2025 **Upadhyay, R.R., Padhye, S.** *[Privacy-Preserving Image Retrieval Based on Threshold Secret Sharing](#)*. **Human-2025 Symposium**.
- 2024 **Upadhyay, R.R., Padhye, S.** *[Efficient and Secure MPC Through Integration of FHE and Proxy Re-Encryption](#)*. **SN Computer Science**.
- 2024 **Upadhyay, R.R., Padhye, S.** *[Advancements in FHE over the Integers: A Comprehensive Survey and Analysis](#)*. **Surveys in Mathematics and its Applications**.
- 2023 **Upadhyay, R.R., Padhye, S.** *[Enhancing FHE Over the Integers: Beyond Binary Numbers and Batch Processing](#)*. **ICCNSML 2023**. Springer LNNS.

2023 **Upadhyay, R.R., Padhye, S.** *Multi-Key Fully Homomorphic Encryption Scheme Over the Integers*. REDCYSEC 2023. Springer LNNS.

Awards & Grants

- June 2025 **Financial Support for PrivCrypt 2025**, Supported by **CIMPA, CRSI, and BOSCH**, Munich, Germany
For presenting the paper "*Towards Privacy and Integrity: SNARK-Driven Verifiable FHE for Outsourced Computation*" at PrivCrypt 2025, a co-located workshop of ACNS 2025.
- March 2025 **Best Paper Award**, 3rd Doctoral Symposium on Human-Centered Computing (Human-2025)
Awarded for the paper "*SoK: Digital Signatures from VOLEitH*".
- March 2025 **Student Stipend (IACR)**, *Real World Crypto Symposium (RWC) 2025*, Sofia, Bulgaria
Awarded by IACR to attend the symposium and engage with the global cryptography community.
- March 2025 **Student Stipend**, *TPMPC 2025*, IISc Bangalore, India
To attend the Theory and Practice of Multi-Party Computation Workshop.
- Sep – Dec 2024 **CIMPA Fellowship / Trimester Laureate**, *Institut Henri Poincaré (IHP)*, Paris, France
Selected as one of three global **Laureates** for the "*Post-Quantum Algebraic Cryptography*" program.
- Dec 2023 **CIMPA Financial Support**, *CIMPA Research School*, Pondicherry University, India
For attending the Research School on Post-Quantum Cryptography.
- 2022 **CSIR-UGC JRF and NET Qualification**, *CSIR, India*
Qualified for Junior Research Fellowship (JRF) and Lectureship (NET).
- 2021, 2022 **GATE Qualified**, *Ministry of Education, India*
Graduate Aptitude Test in Engineering.